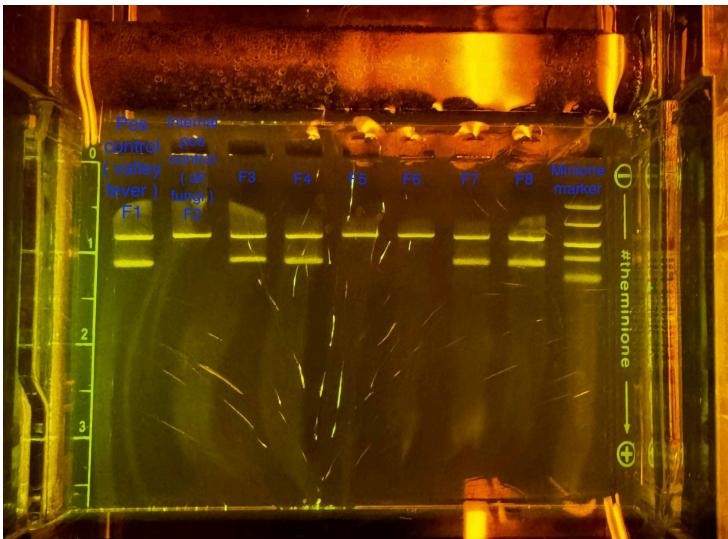


Valley Fever Lab Write Up

Question: Which area(s) have the highest exposure to Valley Fever

Claim: Kern, San Luis Obispo, Fresno have the highest exposure to valley fever. I can make this claim with the evidence provided. I am comparing the evidence given to the other counties and it shows that the counties we have are the highest.

Evidence:



County	Number of Positive Samples	Incidence of VF (Low, Medium, High)
Fresno	5.5 = 5	high
Kern	6.6 = 6	high
Los Angeles	4.4 = 4	med
Merced	4.4 = 4	med
Orange	2.2 = 2	low
Riverside	3.4 = 3.5	med
Sacramento	2.2 = 2	low
San Diego	3.3 = 3	med
San Francisco	1.1 = 1	low
San Luis Obispo	5.5 = 5	high

Reasoning:

The Kern, San Luis Obispo, and Fresno lanes show bands in the same location as the positive control indicating they are positive for valley fever. This suggests the fungus is present in those counties while the

other samples do not match. The bands are distinct and because the gel ran correctly, the results can be considered reliable.

Post Lab Analysis Questions

1. Are your results valid? Explain your reasoning,

Our results were valid because the positive control showed bands and the DNA band is visible so it means the gel ran correctly.

2. Based on the gel electrophoresis data, rank the incidence of Valley Fever in each of the counties (from highest to lowest).

The highest were Kern County, Fresno, and San Luis Obispo. Medium it was Los Angeles, Merced, Riverside, and San Diego County. The lowest were Orange County, Sacramento, and San Francisco.

3. Did the results from the gel electrophoresis match your predictions from the maps in the prelab packet? Why or why not?

The results did match my prediction because the areas that are hotter and drier showed higher amounts of valley fever that matched my predictions.

4. Based on the results of this feasibility study, would you suggest that the local universities, Health Department, and the Centers for Disease Control (CDC) expand this study to the other counties in California, and in other states across the country? Why or why not?

Yes, it would help find more information about high risk areas and better protect people from getting sick.

4. How could the findings from this soil testing lab contribute to public health initiatives and community awareness regarding *Coccidioides*?

It can help by raising awareness, warn people in high risk areas, and help prevent valley fever infections.